

Charles Ran

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AWARDS

ICPC World Finalist 11th Place, ICPC NA Championship	2026
Top 200 (Top 10 in Canada), William Lowell Putnam Mathematical Competition	2024
Master (Top 1% Global), Codeforces Top 1% Global, LeetCode	

EDUCATION

University of British Columbia <i>Bachelor of Science in Computer Science and Mathematics (Honors, Co-op)</i> <i>Master of Management (Concurrent Dual Degree)</i> • 4th Year Standing; 88% Cumulative Average; Dean's List 2024, 2025, 2026.	Vancouver, Canada Sep 2024 – May 2028
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EXPERIENCE

Research Internship in ML Reasoning <i>Noah's Ark Lab, Huawei Canada</i> • Researching deep learning architectures for few-shot reasoning and test-time computation.	May 2026 – Present Markham, Canada
Undergraduate Research in Combinatorics <i>UBC; Supervised by Dr. Gabriel Currier</i> • Investigating Erdos-Ko-Rado-type intersection theorems and their generalizations to finite vector spaces. • Proved novel results , established best-known bounds and conducted literature reviews.	Sep 2025 – Present Vancouver, Canada
Software Developer <i>UBC Competitive Programming Club</i> • Developed new club website icpc.cs.ubc.ca in SvelteKit with professional design practices using ShadCN UI , Figma assets, and TailwindCSS . Leveraged Cloudflare for hosting, DNS services, and security. • Self-hosted backend infrastructure with Docker on a Linux server and populated dynamic content in an SQLite database with S3 cron backups by scraping historical data with Playwright and Cheerio .	Sep 2024 – Present Vancouver, Canada

PROJECTS

CLVSA: Variational Seq2Seq ConvLSTM Model with Attention <i>PyTorch, Pandas, Scikit-Learn, Hugging Face</i> • Implemented and extended Wang et al. (2019)'s CLVSA model for financial time-series prediction . Wrote a reproducibility report on the results, analyzing model performance and limitations. • Built a Convolutional LSTM sequence-to-sequence architecture to decode future market-movement labels. • Implemented self-attention and inter-attention between encoder and decoder states, and a variational component using stochastic latent states, backward decoding, and KL divergence regularization. • Built a customizable training pipeline and evaluated the model on OHLCV equities and futures data.	Mar 2026 – Apr 2026 rancharles/clvsa
Virtual Olympiad Platform <i>TypeScript, SvelteKit, React.js, WebSocket, PostgreSQL, Docker</i> • Innovated an educational collection of platforms, databases, and tools for math and science competitions. • Designed dynamic, responsive, frontends using SvelteKit , React.js , TailwindCSS , and Figma . • Built a web scraper and parser for online math with Cheerio to populate a PostgreSQL database. • Developed a real-time multiplayer math competition game server and APIs through WebSocket , Express.js and Node.js , integrated with a self-hosted Supabase for Google OAuth with PKCE flow. • Self-hosted backend infrastructure using Linux cloud servers and Docker for scalability and redundancy.	Sep 2022 – Aug 2025 volympiad.org · virtual-olympiad

TECHNICAL STACK

Languages: Python, JavaScript, TypeScript, C, C++, Java, Julia, Assembly, Bash, GLSL, SQL
Machine Learning, Data Science: PyTorch, NumPy, Pandas, Scikit-learn, Jupyter, OpenCV
Web Development: HTML, CSS, SvelteKit, React.js, TailwindCSS, PostgreSQL, MongoDB, Node.js, Express.js, WebSocket, Docker, Cloudflare, Figma, Git, Linux, OpenGL, WebGL, Supabase, Firebase